

Algebra II with Trigonometry

Chapter 11.1

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:
[4652951_alg-2trig-h-chp-11-1-note-docx.pdf](#)

Problems

1. The circle with center at $(7, -5)$ passes through the origin. What is the sum of all possible radii that are integers?
2. The parabola $y^2 = 8x$ and the circle $x^2 + y^2 = 16$ intersect at points A and B in the first quadrant. Find the exact distance between A and B.
3. Let $x^2 = 4py$ be a parabola whose latus rectum has length 10. The focus F is at $(0, p)$. A line passing through F is tangent to the parabola. What is the absolute value of the slope of this tangent line?
4. A parabola with vertex at the origin opens upward and has a directrix at $y = -d$. If the area enclosed between the parabola and the x-axis from $x = -d$ to $x = d$ is exactly 8, what is d ?
5. Given two parabolas, $y^2 = 8x$ and $y^2 = 8(x - 6)$, determine the x-coordinate of their intersection point(s) with the smallest possible positive value.
6. A point P moves in the plane so that its distance from $(3, 0)$ is equal to its distance from the line $x = -1$. Find the equation of the locus, and then determine the coordinates where this locus meets the x-axis.